FX Momentum and Value Strategy using long-span data

Empirical Analysis on a Cross-Sectional Dataset (Refinitiv, Bloomberg)

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Abstract. *This paper investigates two systematic cross-sectional risk premia strategies in foreign exchange markets: a short-term momentum strategy and a Purchasing Power Parity (PPP)-based value strategy. Using daily spot rates and monthly one-month forward rates from Reuters (1998–2024), we construct five fixed-composition long/short portfolios under a four-day lookback, one-month-holding momentum framework, delivering Sharpe ratios of up to 0.44. For the value strategy, restricting to G10 currencies and applying quarterly rebalancing delivers a Sharpe ratio of 0.38 in-sample (1998–2015), while extending to the full cross-section of 21 currencies produces systematically negative performance – attributed to the slower PPP mean-reversion dynamics outside the G10. Out-of-sample (2015–2026), the G10 quarterly strategy is evaluated using bid-ask adjusted excess returns from Refinitiv 3M forwards, accounting for transaction costs. A Balassa–Samuelson productivity-adjusted signal currently yields poor results, motivating further refinement.*

1 Introduction

Currency markets exhibit well-documented departures from the efficient markets hypothesis. Among these, momentum and value anomalies provide persistent risk premia that cannot easily be explained by standard factors and form the basis of widely-used systematic FX investment strategies.

Momentum in currency markets exploits the tendency of recently appreciating currencies to continue doing so, arising from the heterogeneity of market participants (Hong and Stein, 1999). Menkhoff et al. (2012) provided the canonical FX treatment, showing that currency momentum earns a significant risk premium attributable to global FX volatility risk. Post-2012, raw momentum profitability has deteriorated markedly (Bartel et al., 2025; Iwanaga and Sakemoto, 2025), consistent with compressed interest rate differentials and heightened central bank intervention suppressing directional trends.

Value in currency markets is rooted in the long-run tendency of real exchange rates to revert to their PPP equilibrium. While PPP famously fails at short horizons (Rogoff, 1996), convergence is well-documented over three-to-five-year horizons for developed economies (Frenkel, 1986; Lothian and Taylor, 1996; Mark, 2001). Menkhoff et al. (2017) confirm that the currency value premium is robust, particularly for G10 currencies. Asness, Moskowitz, and Pedersen (2013) demonstrate that value and momentum carry a persistent negative correlation of approximately -0.40 across asset classes, making them natural complements. The Deutsche Bank Currency Returns (DBCR) index operationalises both in investable cross-sectional benchmarks (Deutsche Bank, 2007).

We aim to expand this analysis using long-span data from 1998 to 2026 looking for recent trends of these two historical systematic trading strategies.

2 Momentum Strategy

2.1 Strategy Construction

Assuming CIP holds, the excess log return of a long (resp. short) position in currency i, entered via a one-month forward F_t and exited at the subsequent spot S_{t+1}, is:

$$r_{t+1}^{\text{long}} = \ln\left(\frac{S_{t+1}}{F_t}\right), \qquad r_{t+1}^{\text{short}} = \ln\left(\frac{F_t}{S_{t+1}}\right). \tag{1}$$

Log returns are used throughout: they are additive across time, consistent with the multi-period nature of the strategy, and ensure exact symmetry between the long and short legs of each portfolio. This also allows comparison with the existing literature.

In this paper, the momentum signal is the sum of daily log spot returns over the four calendar days immediately preceding the rebalancing date. Portfolios are rebalanced on the first trading day of each month, coinciding with the availability of new forward rates. A currency receives a ‘Long’ signal if its four-day cumulative return is positive, ‘Short’ otherwise.

We construct five fixed-composition portfolios of five currencies each (Table 1), plus a dynamic Top-3/Bottom-3 portfolio that at each date goes long the three highest-momentum and short the three lowest-momentum currencies from the full cross-section.

Table 1: Fixed-Composition Portfolio Constituents

Portfolio	Currencies
Portfolio 1	AUSTDOL, BRACRUZ, BULGLEV, CNDOLLR, CHILPES
Portfolio 2	CROATKN, USEURSP, HUNFORT, INDRUPE, INDORUP
Portfolio 3	ISRSHEK, JAPAYEN, MEXPESO, NZDOLLR, NORKRON
Portfolio 4	PHILPES, POLZLOT, CISRUBM, SINGDOL, COMRAND
Portfolio 5	SWEKRON, SWISSFR, THABAHT, CZECHCM, GBP
Top-3 / Bottom-3	Dynamic (full cross-section, ranked monthly)

Note: Six currencies have missing forward data at the start of the sample and are dynamically excluded from the momentum ranking during those periods: BRACRUZ before August 2000; BULGLEV, CHILPES, CROATKN, ISRSHEK, and CISRUBM before April 2004. This reflects delayed forward rate availability in the Reuters dataset. They remain in their fixed portfolio slots but contribute zero return during those periods.

2.2 Performance Results and Moments Analysis

Table 2 reports full-sample annualised performance metrics (February 1998 – October 2024, 321 monthly observations).

Table 2: Annualised Performance Metrics – Full Sample (1998–2024)

Portfolio	Ann. Ret. (%)	Ann. Vol. (%)	Sharpe	Skewness	Exc. Kurtosis
Portfolio 1	7.03	33.52	0.210	+0.958	8.043
Portfolio 2	16.63	37.89	0.439	+0.320	2.648
Portfolio 3	1.55	28.74	0.054	+0.372	6.162
Portfolio 4	13.31	31.30	0.425	−0.343	4.054
Portfolio 5	3.43	33.13	0.103	+0.439	1.312
Top-3/Bot-3	5.03	32.92	0.153	+0.254	3.232

Sharpe $= 12\bar{r} \,/\, (\sqrt{12}\,\hat{\sigma})$. Skewness and excess kurtosis via `scipy.stats` (Fisher convention; normal = 0).

Positive cross-sectional risk premia. All six portfolios deliver positive Sharpe ratios, confirming the cross-sectional FX momentum premium (Menkhoff et al., 2012).

Positive skewness. Four out of five fixed portfolios display positive skewness, contrasting sharply with carry trade crash risk (Brunnermeier et al., 2009). Portfolio 4 (PHILPES, CISRUBM, COMRAND) is the exception.

Fat tails. Excess kurtosis is uniformly positive. Portfolio 1 peaks at 8.04, driven by episodic large moves in BRACRUZ and BULGLEV.

2.3 Return Distributions

Figure 1 shows the fitted return distributions of the five fixed portfolios. All display leptokurtic profiles with heavier tails than the normal benchmark. Figure 2 shows the Top-3/Bottom-3 distribution with skewness +0.25 and excess kurtosis 3.23, clearly fat-tailed.

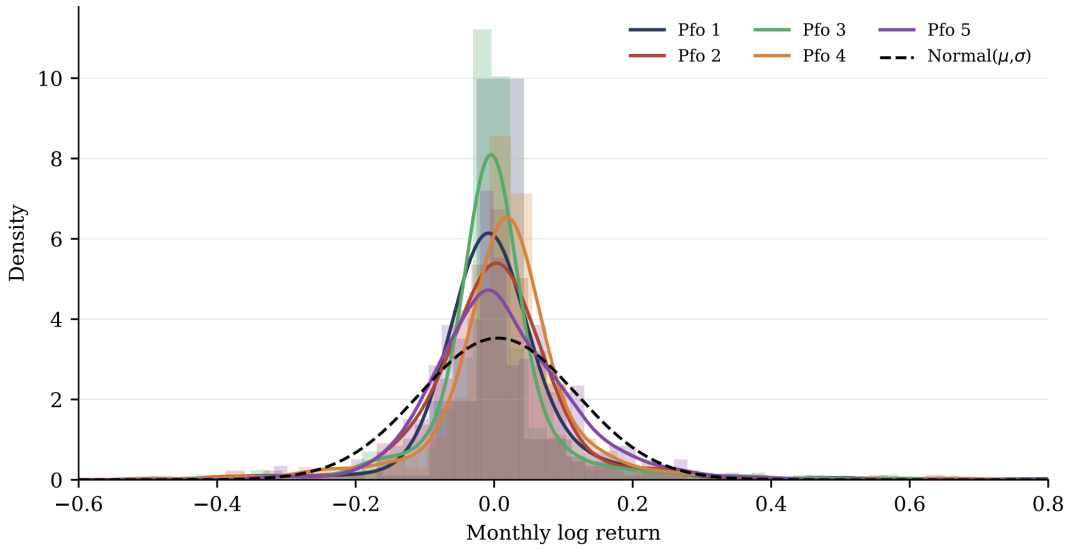

Figure 1: Return distributions of the five fixed portfolios (kernel density estimates overlaid on histograms, density scale). Dashed line: normal fitted to pooled returns. All portfolios display excess kurtosis; four out of five show positive skewness.

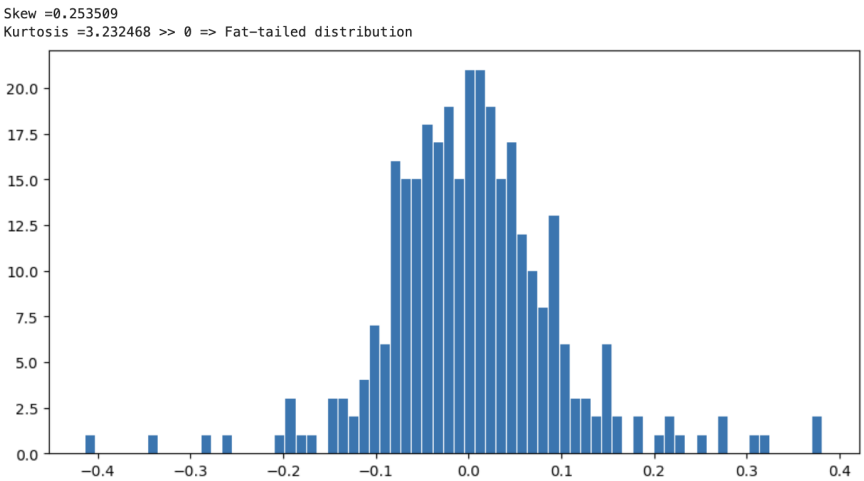

Figure 2: Distribution of monthly returns for the dynamic Top-3/Bottom-3 portfolio (321 observations). Skewness = +0.25; excess kurtosis = 3.23 $\gg$ 0, confirming a fat-tailed distribution consistent with momentum capturing occasional large positive directional moves.

2.4 Cumulative Performance

The post-2012 structural decay in momentum profitability is consistent with Bartel et al. (2025): compressed interest rate differentials and increased central bank intervention suppressed the directional trends that momentum relies on. The dynamic Top-3/Bottom-3 portfolio shows

greater resilience, as full-universe aggregation diversifies idiosyncratic reversals (find total returns series in the corresponding github repository).

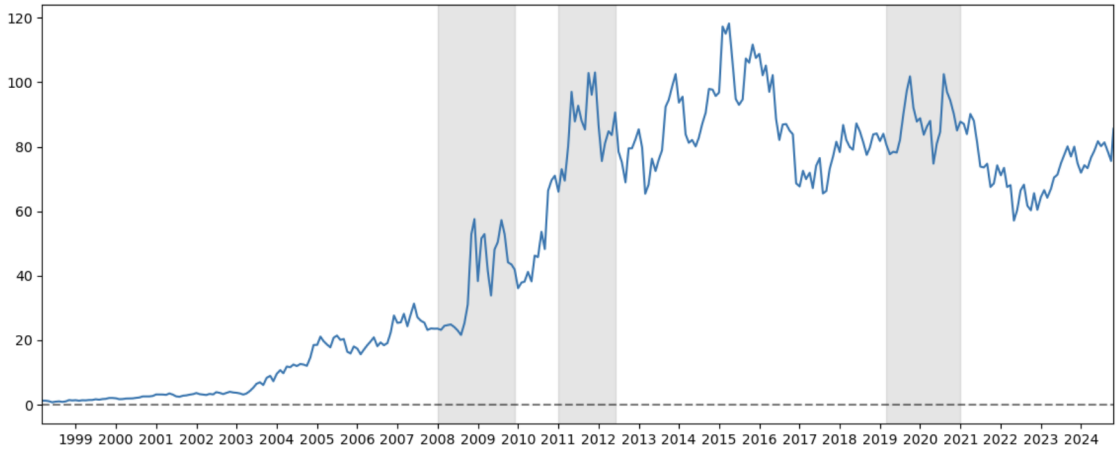

Figure 3: Cumulative return of Portfolio 2 (CROATKN, USEURSP, HUNFORT, INDRUPE, INDORUP). The portfolio achieves the highest annualised return of all fixed portfolios (16.63%, Sharpe = 0.44) but also the highest volatility (37.89%), reflecting compounding of strong momentum signals in central/eastern European and Asian currencies over the full sample.

2.5 Next Steps

Two extensions are in progress: (i) incorporating transaction costs via the bid-ask spreads available in the Reuters dataset – essential for net profitability assessment, particularly in less liquid emerging market pairs; and (ii) computing the full excess return $\ln(S_{t+1}/F_t)$, which under CIP already embeds the interest rate differential, rather than using spot returns only for the signal.

This extension has, however, been addressed in the Value Strategy.

3 PPP-Based Value Strategy

3.1 Strategy Construction

Define the real exchange rate of currency i (direct quote, USD per foreign unit) as:

$$Q_{i,t} = S_{i,t} \times \frac{P_{f,i}}{P_{\text{USD}}}, \tag{2}$$

where $P_{f,i}/P_{\text{USD}}$ is the OECD PPP conversion factor for GDP (national currency per USD). The cross-sectional signal is:

- $Q_{i,t} > 1$: overvalued in real terms $\rightarrow$ **short**
- $Q_{i,t} < 1$: undervalued in real terms $\rightarrow$ **long**

Weights are equal ($1/n$ per leg) and lagged one period to prevent look-ahead bias. The return for period t is the log change in the spot rate $\ln(S_{t+1}/S_t)$, computed at the rebalancing frequency. We test two rebalancing frequencies (monthly and quarterly) and two currency universes: (i) the

full cross-section (21 currencies with available OECD PPP data) and (ii) the restricted G10 (AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK).

Data note. CROATKN and BULGLEV are excluded from the value strategy: both countries have adopted or are joining the Euro (Croatia in 2023, Bulgaria scheduled for 2025), rendering their bilateral USD PPP series economically irrelevant for a forward-looking signal. PHILPES and THABAHT are excluded due to absent OECD PPP coverage.

3.2 In-Sample Results (1998–2015)

3.2.1 Full Universe

Table 3 reports the in-sample performance for the full 21-currency universe. All three specifications deliver negative Sharpe ratios, reflecting the inability of annual OECD PPP data to serve as a reliable short-to-medium-term signal for emerging market and frontier currencies. The descriptive statistics for the monthly in-sample return series confirm the problem: the median return is zero, and the distribution is heavily compressed around zero, indicating that most currencies contribute nothing informative to the cross-sectional ranking on most rebalancing dates.

Table 3: Full-Universe Value Strategy – In-Sample Performance (1998–2015)

Specification	Sharpe	Mean (monthly)	Std	Obs.
Monthly rebalancing	−0.030	−0.000008	0.000897	4,696
Quarterly rebalancing	−0.359	–	–	–
Quarterly (median date)	−0.388	–	–	–

Sharpe annualised (×12 monthly; ×4 quarterly). The monthly return series has min = −0.023, max = +0.014, with the 25th, 50th, and 75th percentiles all equal to zero – indicating most rebalancing dates contribute no signal.

3.2.2 G10 Universe

Restricting to G10 currencies dramatically reverses the picture. Table 4 reports the G10 in-sample results.

Table 4: G10 Value Strategy – In-Sample Performance (1998–2015)

Universe	Rebal.	Sharpe	Mean	Std	Min	Max	Obs.
G10	Monthly	0.317	0.198%	2.169%	−5.91%	8.63%	216
G10	Quarterly	**0.381**	0.702%	3.683%	−8.59%	8.67%	72

Sharpe annualised (×12 monthly; ×4 quarterly). Mean and std are per-period (not annualised). Skewness: quarterly = -0.021, monthly = -0.099. Excess kurtosis: quarterly = -0.276, monthly = +0.515. The near-zero skewness and slightly negative excess kurtosis for the quarterly G10 strategy indicate a near-Gaussian return profile.

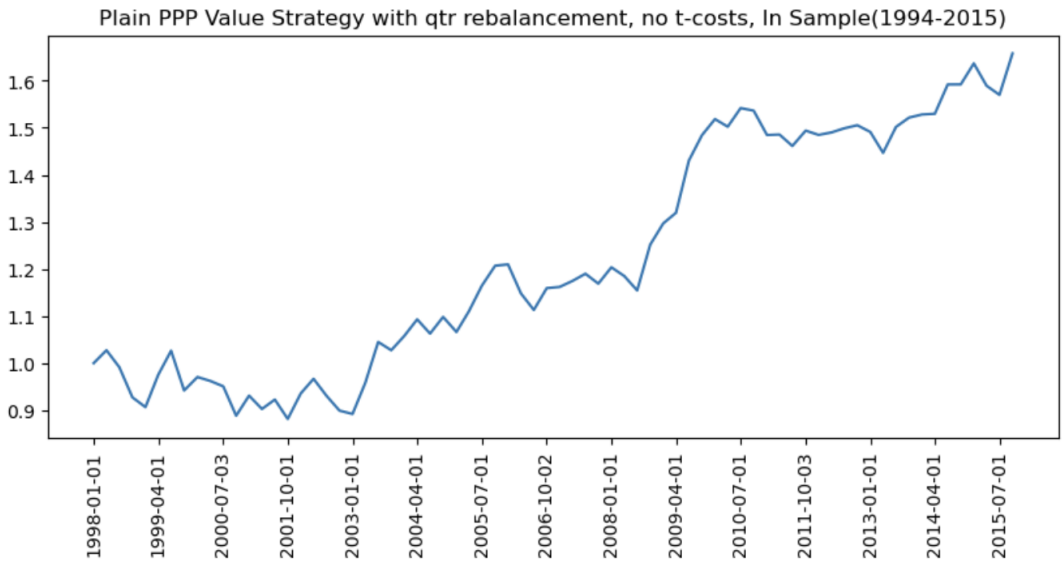

Figure 4: Cumulative return of the G10 PPP value strategy (quarterly rebalancing, no transaction costs), in-sample 1998–2015. The strategy reaches approximately 1.6× its initial value by 2015.

3.3 The G10 Restriction: A Theoretical Prediction

The reversal between full-universe (negative) and G10 (positive) Sharpe ratios carries a clear theoretical interpretation. PPP mean reversion operates through trade and capital flows correcting real exchange rate misalignments. For G10 currencies, this is well-documented with half-lives of three to five years (Frenkel, 1986; Lothian and Taylor, 1996), providing a reliable medium-term anchor. For less liquid markets, capital controls, structural inflation differentials, and political risk episodes – e.g. the Brazilian Real and Russian Rouble – produce large one-sided devaluations that overwhelm the PPP signal.

The superiority of quarterly over monthly rebalancing for G10 is also consistent with theory: monthly rebalancing is too frequent relative to multi-year PPP half-lives, whereas quarterly horizons allow partial adjustment to materialise (Mark, 2001). The DBCR Valuation Index reaches the same conclusion (Deutsche Bank, 2007).

3.4 Out-of-Sample Evaluation (2015–2026)

The out-of-sample period (January 2015 – March 2026) uses bid-ask adjusted excess returns computed from Refinitiv LSEG 3-month forward rates to incorporate transaction costs explicitly. For a long position in currency i:

$$r_{t+1}^{\text{long, net}} = \ln\left(\frac{S_{t+1}^{\text{bid}}}{F_t^{\text{ask}}}\right), \qquad r_{t+1}^{\text{short, net}} = \ln\left(\frac{F_t^{\text{bid}}}{S_{t+1}^{\text{ask}}}\right), \tag{3}$$

where the bid is used on the favourable side and the ask on the unfavourable side, correctly penalising round-trip transaction costs. The portfolio weights $w_{i,t}$ are carried forward from the in-sample signal construction using the same G10 quarterly rebalancing rule applied recursively on the full dataset.

The G10 PPP value strategy achieves a Sharpe ratio of 0.231, with a mean monthly return of 0.16% and volatility of 1.38%. The return distribution is right-skewed (skewness = +0.429) and fat-tailed (excess kurtosis = 5.817), with 108 observations ranging from -5.465% to +5.455%. This suggests the strategy's resilience to transaction costs, despite occasional large positive or negative moves.

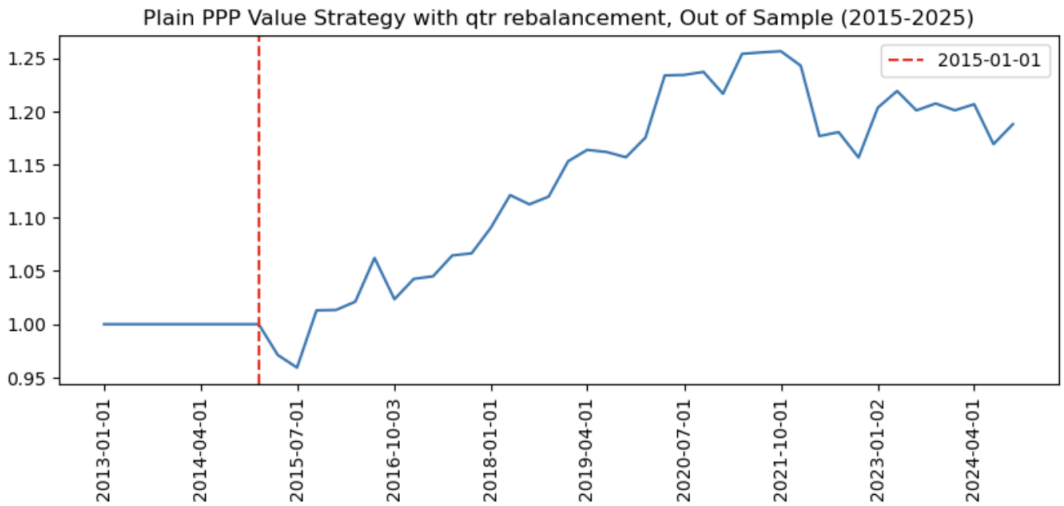

Figure 5: Cumulative return of the G10 PPP value strategy (quarterly rebalancing), out-of-sample 2015–2026, net of transaction costs using Refinitiv bid-ask adjusted 3M forward rates.

3.5 Balassa–Samuelson Extension

Standard PPP ignores that richer economies display structurally higher price levels (Balassa, 1964; Samuelson, 1964). We attempt to correct for this using World Bank GDP per capita in current USD as a productivity proxy. Initial results are negative and economically insignificant, worse than the unadjusted G10 strategy. This remains a key open avenue for future research.

4 Strategic Implications

The post-2012 decay in raw momentum and the moderate standalone Sharpe ratios for the value strategy suggest that neither factor alone provides a robust cross-sectional investment solution in the modern FX landscape. Asness, Moskowitz, and Pedersen (2013) document a persistent -0.40 correlation between value and momentum, arising because momentum exploits mid-term behavioural underreaction while value profits from the eventual fundamental correction. When a currency overshoots its PPP fair value due to a momentum trend, the value strategy builds a counter-position, providing a natural hedge against momentum crashes (Daniel and Moskowitz, 2016). Combining both signals into a multi-factor cross-sectional portfolio is the natural next step, expected to deliver Sharpe ratios materially above either factor alone.

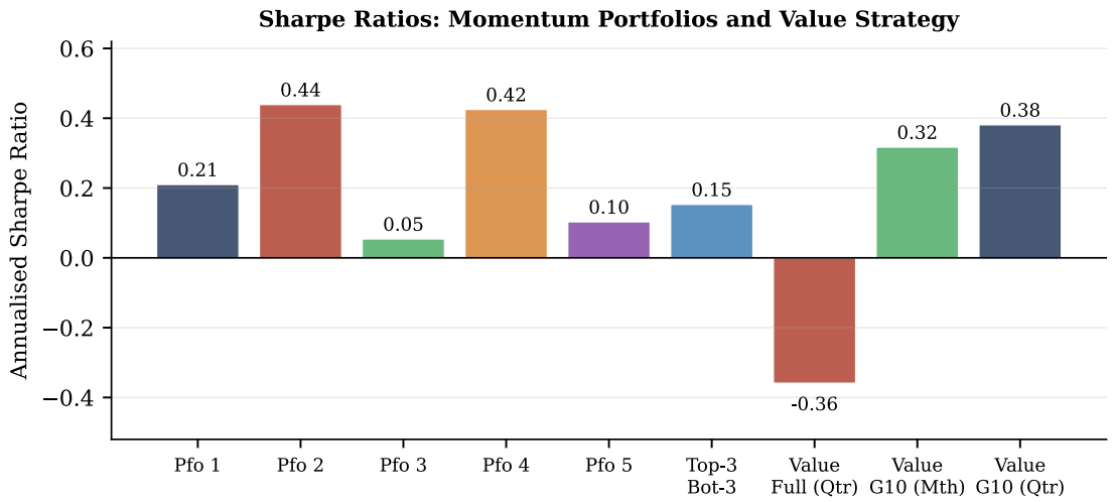

Figure 6: Annualised Sharpe ratios across all strategies. Left six bars: momentum portfolios. Right three bars: value strategy specifications. The negative bar (full-universe quarterly) illustrates the cost of non-G10 currencies in the value framework. G10 quarterly value (rightmost) achieves an in-sample Sharpe of 0.38.

5 Data

Exchange rates (in-sample). Daily spot rates and monthly one-month forward rates from the Reuters WM/Reuters dataset (1 January 1998 – 23 October 2024; 6,995 daily observations; 321 monthly forward observations per series). All series are converted to USD per unit of foreign currency. USEURSP and GBP/USD require no inversion. CROATKN and BULGLEV forward rates are quoted against EUR and require conversion via the EUR/USD spot on each rebalancing date. Mid-rates are used throughout.

Exchange rates (out-of-sample). Daily spot bid/ask and 3-month forward bid/ask rates for the G10 universe (AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK) sourced from Refinitiv LSEG Workspace via the `lseg.data` Python package (RICs: `CCY=` for spot, `CCY3MV=` for 3M forward outrights), covering 1 January 2015 – 31 March 2026. Bid-ask spreads are used to compute net excess returns as in equation (3).

PPP data. Annual GDP-based PPP conversion factors from the OECD Data Explorer (national currency per USD), forward-filled to rebalancing dates. data-explorer.oecd.org. PHILPES and THABAHT excluded (no OECD coverage); CROATKN and BULGLEV excluded (Euro Area accession).

GDP per capita. World Bank Open Data, GDP per capita in current USD, annual, used for the Balassa–Samuelson adjustment. data.worldbank.org

Conclusion

This paper documents positive cross-sectional FX momentum risk premia with Sharpe ratios up to 0.44 over a 26-year sample. Predominantly positive skewness distinguishes momentum from carry's crash risk, and the post-2012 structural decay is confirmed. For the PPP value strategy, the G10 restriction is theoretically motivated: PPP reversion requires arbitrage mechanisms

that operate far more slowly outside developed currency markets. The G10 quarterly strategy achieves a Sharpe of 0.38 in-sample. Out-of-sample evaluation using bid-ask adjusted Refinitiv 3M forward returns is underway, with a data-alignment fix being finalised. The Balassa–Samuelson adjustment remains an open problem. The immediate research agenda is to: (i) produce final OOS Sharpe ratios; (ii) incorporate bid-ask adjusted transaction costs in the momentum backtest; and (iii) combine value and momentum into a multi-factor cross-sectional strategy.

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